

# Patrick Sherlock

psher@mit.edu | patrickjsherlock01@gmail.com | <https://patricksherlock.github.io/>

---

## Education

2019 - 2023      University of Southern California  
B.S. Neuroscience, Minor Linguistics

---

## Research Positions

2025 - present      Technical Research Associate, Massachusetts Institute of Technology  
*Advisor: Dr. Evelina Fedorenko*

2023 - 2025      Project Coordinator, Vanderbilt University  
*Advisor: Dr. Sarah Brown-Schmidt*

2022 - 2023      Undergraduate Research Assistant, University of Southern California  
*Advisor: Dr. Zuzanna Fuchs*

2020 - 2023      Undergraduate Research Assistant, University of Southern California  
*Advisors: Dr. Kirsten Lynch, Dr. Jeiran Choupan*

---

## Publications

*In press*      Lord, K., Kekes-Szabo, S., **Sherlock, P.**, Duff, M.C., & Brown-Schmidt, S. Moderate-severe traumatic brain injury disrupts core mechanisms of online language processing and use. *Cortex*.

*In press*      Casto, C., Poliak, M., Tuckute, G., Small, H., **Sherlock, P.**, Wolna, A., Lipkin, B., D'Mello, A. M., & Fedorenko, E. The cerebellar components of the human language network. *Neuron*.

2023      Lan, H., Lynch, K., Custer, R., Shih, N., **Sherlock, P.**, Sepehrband, F., Toga, A., & Choupan, J. Weakly supervised perivascular spaces segmentation with salient guidance of Frangi filter. *Magnetic Resonance in Medicine*, 89, 6, 2419-2431.

---

## Manuscripts

*In preparation*      **Sherlock, P.**, Duff, M.C., & Brown-Schmidt, S. Moderate-severe traumatic brain injury disrupts online grammatical processing.

*In preparation*      **Sherlock, P.**, Duff, M.C., & Brown-Schmidt, S. Moderate-severe traumatic brain injury disrupts online referential processing.

*In preparation* Lord, K., Zhou, Y., **Sherlock, P.**, & Brown-Schmidt, S. Different demands, different memory: Conversational role and memory for the contexts of language use.

---

### Conference Abstracts

- 2025 **Sherlock, P.**, Duff, M.C., & Brown-Schmidt, S. The impact of TBI on the use of referential context in syntactic ambiguity resolution. *Human Sentence Processing*. College Park, MD, USA. 27-29, March 2025. [Poster]
- 2024 Volante, C., Diachek, E., **Sherlock, P.**, & Brown-Schmidt, S. Referential modification and memory for object features. *Human Sentence Processing*. Ann Arbor, MI, USA. 16-18, May 2024. [Poster]
- 2022 **Sherlock, P.**, Lynch, K., Choupan, J., & Toga, A. Anterior and middle cerebral artery caliber and PVS size in young adults. *Organization for Human Brain Mapping*. Glasgow, Scotland, UK. 19-23, June 2022. [Poster]
- 

### Guest Lectures

- 2024 Applying to Postbac Research Positions  
*Summer Professional Development Series for Undergraduate Trainees*  
Facilitator: Dr. Whitney Barnett, Vanderbilt University
- 2024 Language and Conversational Memory in Law and Policy  
*PSY-PC 7500: Advanced Topical Seminar: Cognition in the Real World*  
Instructor: Dr. Lisa Fazio, Vanderbilt University
- 

### Additional Training

- 2022 NASSLLI  
Volunteer and attendee, University of Southern California
- 

### Outreach

- 2026 “Language in the Mind and Brain” Course  
*MIT Educational Studies Program HSSP, Cambridge, MA*
- 2026 Field Trip Volunteer  
*MIT Museum, Cambridge, MA*

|      |                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Cambridge School Volunteers NetPal<br><i>Cambridge Street Upper School, Cambridge, MA</i>                                          |
| 2025 | “What it Means to be Human” Course<br><i>MIT Educational Studies Program Splash, Cambridge, MA</i>                                 |
| 2024 | “The Science of Language and Bilingualism” Science Fair Booth<br><i>Glendale Spanish Immersion Elementary School, Oak Hill, TN</i> |

---

**Awards**

|      |                                                  |
|------|--------------------------------------------------|
| 2025 | McGovern Institute for Brain Research SPOT Award |
|------|--------------------------------------------------|

---

**Technical Proficiency**

|                 |                                                                  |
|-----------------|------------------------------------------------------------------|
| Coding          | Python (PsychoPy, PyLink, Pandas), Bash, GitHub, SLURM, MATLAB   |
| Methodologies   | MRI, Eye-Tracking                                                |
| Data Collection | jsPsych, Prolific, Qualtrics, REDCap, SONA, Gorilla, PCIBex      |
| Analysis Tools  | SPM, CONN, R                                                     |
| Applications    | Jupyter, Praat, Microsoft Office Suite, Google Workspaces, Gephi |

---

**Natural Languages**

|            |                             |
|------------|-----------------------------|
| English    | Native                      |
| Spanish    | Full working proficiency    |
| Portuguese | Limited working proficiency |
| Italian    | Basic                       |